BRUCE™

Bruce Best Bid and Offer (BBBO) Specification

Nasdaq

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Overview

The Bruce Best Bid and Offer (BBBO) is a direct data feed product offered by *Bruce ATS*, providing the platform's best bid and offer position for securities traded.

- **Quotation Messages:** The BBBO will provide the current inside, on a per symbol basis, for the orders posted on the Bruce trading system. This will be done using the BBBO's Quotation Message, which provides the symbol, the best bid price, the best ask price, and the cumulative shares available at the indicated prices.
- **Administrative Messages**:
  - *Stock Directory* - provide basic security data for the available symbols
  - *Trading Action* - inform market participants when a security is halted or released for trading
  - *Reg SHO Restriction* - provide the current RegSHO short sell restrictions being enforced
- **Event Controls:** Used to communicate key market events on Bruce ATS including start and end of day.

Architecture

BBBO will be made up of a series of sequenced messages. Each message is variable in length based on the message type. The messages that make up the BBBO data feed are typically delivered using a higher-level protocol that takes care of sequencing and delivery guarantees.

BRUCE ATS offers the BBBO data feed in two delivery protocol options:

- SoupBinTCP
- MoldUDP64

In the market data messages, instruments are identified by a stock locate code – a low lying integer employed with the intent of serving as an array index for rapidly looking up instrument details. The locate codes are dynamically assigned each day, starting with a value of 1, and communicated via the Stock Directory message. An instrument's locate code will not change intraday. However, there should be no expectation that the assignment will remain the same across multiple days. The Stock Locate code appears in all messages, and at the same position in all messages to support efficient filtering. A default value of 0 will be applied to messages which are not stock dependent.

Data Types

All integer fields are big-endian (network byte order) binary encoded numbers. They are unsigned, unless otherwise noted.

All alphanumeric fields are ASCII fields which are left-justified and padded on the right with spaces.
Prices are integer fields, supplied with an associated precision. When converted to a decimal format, prices are in fixed point format, where the precision defines the number of decimal places. For example, a field flagged as Price (4) has an implied 4 decimal digits.
Timestamps reflect the system time when the outbound message was generated and are represented as the number of nanoseconds since the start of the unix epoch (January 1, 1970).

Message Formats

System Event Message

The System Event message is used to communicate key market or data feed control events.

SYSTEM EVENT MESSAGE				
Name	**Offset**	**Length**	**Value**	**Notes**
Message Type	0	1	"S"	System Event
Stock Locate	1	2	Integer	Always 0
Time Stamp	3	8	Timestamp	Time stamp
Event Code	11	1	Alphanumeric	Denotes the type of event for which the message is being generated. See list below for allowable values.

On a daily basis, BBBO will support the following event codes:

SYSTEM EVENT CODES – DAILY	
Code	***Explanation***
O	*Start of Transmissions*: Denotes that BBBO has started its daily transmission schedule
S	*Start of System Hours*: Denotes that Bruce ATS is open and ready to start accepting orders.
Q	*Start of Market Hours:* Denotes that Market Hours orders are available for execution.
M	*End of Market Hours:* Denotes that Market Hours orders are no longer available for execution.
E	*End of System Hours:* Denotes that Bruce ATS is now closed and will not accept any new orders today.
C	*End of Transmissions:* Denotes that BBBO has completed its daily transmission schedule.

Administrative Messages

Stock Directory

At the start of each trading day, Bruce disseminates stock directory messages for all active symbols in its trading system.

STOCK DIRECTORY				
Name	*Offset*	*Length*	*Value*	*Notes*
Message Type	0	1	"R"	Stock Directory Message
Stock Locate	1	2	Integer	Locate Code uniquely assigned to the security symbol for the day
Timestamp	3	8	Timestamp	Time at which the directory message was generated. Refer to Data Types for field processing notes.
Stock	11	8	Alphanumeric	The security symbol for the issue, in Nasdaq symbology. See Appendix B for details.
Market Category	19	1	Alphanumeric	Indicates Listing market for the issue A = NYSE American N = NYSE P = NYSE ARCA Q = NASDAQ V = Investors' Exchange LLC Z = CBOE BZX Exchange <space> = Not Available
Round Lot Size	20	4	Integer	Indicates the number of shares that represent a round lot for the issue
Authenticity	24	1	Alphanumeric	Flags whether the issue is treated by BRUCE as a live/production symbol or a test symbol P = Live/Production T = Test

Stock Trading Action

Bruce ATS uses this administrative message to indicate the current trading status of a security to the trading community.

Prior to the start of market hours, Bruce ATS will send out a Trading Action spin in which the BBBO will deliver a Stock Trading Action message with a *Trading State* of "T" for all securities that are eligible for trading at the start of the market hours. If a security is absent from the pre-opening Trading Action spin, firms should assume that the security is being treated as halted in the Bruce ATS platform.

Afterwards, Bruce ATS will use the Trading Action message to relay changes in trading status for an individual security. Messages will be sent when a stock is:

- Halted/Paused
- Released for trading

STOCK TRADING ACTION MESSAGE				
Name	**Offset**	**Length**	**Value**	**Notes**
Message Type	0	1	"H"	Trading Action
Stock Locate	1	2	Integer	Locate code identifying the security
Time Stamp	3	8	Timestamp	Timestamp
Stock	11	8	Alphanumeric	The security symbol for the issue, in Nasdaq symbology. See Appendix B for details.
Trading State	19	1	Alphanumeric	The current trading state for the issue. The allowable values are: H = Halted / Paused in BRUCE T = Trading

Reg SHO Short Sale Price Test Restricted Indicator

In November 2010, the Securities and Exchange Commission (SEC) implemented changes to Rule 201 of the Regulation SHO (Reg SHO). For details, please refer to SEC Release Number 34-61595. In association with the Reg SHO rule change, Nasdaq introduced the following Reg SHO Short Sale Price Test Restricted Indicator message format.

For all exchange-listed issues, Bruce ATS relays the Reg SHO Short Sale Price Test Restricted Indicator message when it receives an update from the primary listing exchange.
Bruce ATS processes orders based on the most Reg SHO Restriction status value.

REG SHO RESTRICTION				
Name	***Offset***	***Length***	***Value***	***Notes***
Message Type	0	1	"Y"	Reg SHO Short Sale Price Test Restricted Indicator
Stock Locate	1	2	Integer	Locate code identifying the security
Timestamp	3	8	Timestamp	Timestamp
Stock	11	8	Alphanumeric	The security symbol for the issue, in Nasdaq symbology. See Appendix B for details.

REG SHO RESTRICTION				
Name	*Offset*	*Length*	*Value*	*Notes*
Reg SHO Action	19	1	Alphanumeric	Denotes the Reg SHO Short Sale Price Test Restriction status for the issue at the time of the message dissemination. Allowable values are: "0" = No price test in place "1" = Reg SHO Short Sale Price Test Restriction in effect due to an intra- day price drop in security "2" = Reg SHO Short Sale Price Test Restriction remains in effect

Quotation Message

BBBO will broadcast a real-time update every time that the system's best bid and offer quote is updated during the trading day.

BBBO QUOTATION MESSAGE				
Name	Offset	Length	Value	Notes
Message Type	0	1	"Q"	Quotation Message
Stock Locate	1	2	Integer	Locate code identifying the security
Timestamp	3	8	Timestamp	Timestamp
Stock	11	8	Alphanumeric	The security symbol for the issue, in Nasdaq symbology. See Appendix B for details.
Best Bid Price	19	8	Price (4)	The current best bid price for the buy orders posted in BRUCE
Best Bid Size	27	4	Integer	Aggregated number of shares available for buy orders posted on the BRUCE Best Bid Price
Best Offer Price	31	8	Price (4)	The current best offer price for the sell orders posted in BRUCE
Best Offer Size	39	4	Integer	Aggregated number of shares available for sell orders posted on the BRUCE Best Offer Price

Contact Information

Questions about the Nasdaq Basic entitlement and BBBO specification document may be directed to Clientsuccess@nasdaq.com.

Questions about BBBO transmissions may also be directed to Nasdaq Operations Center.

Appendices

Appendix A - Documentation Version Control Log

Version	Date Approved	Comments
1.0	2024-07-30	Initial Version
1.1	2024-08-02	Minor Corrections - added description of stock locate to the Architecture section, and added the stock locate to the Quotation message. Modified some wording to be consistent with BBBO documentation.
3.3	2026-08-11	Minor edits for clarity. Changed Nomenclature from BOSS to Bruce ATS in numerous places. The version number jump is to keep it in line with other market data specs.

Appendix B – Stock Symbol Convention

For all exchange-listed issues, BRUCE currently restricts its symbol length to a maximum of 8 characters, using the Nasdaq symbology for representing the root symbol plus any additional suffixes. For details on the symbology, please refer to the Ticker Symbol Convention page on the Nasdaq Trader website.